



PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 3

Introduction to OOPS Programming

SESSION / ASSIGNMENT – 4

TASKS :

1. Create a base class called `SocialMediaUser` with properties `username` and `followers`, and a method `displayProfile()` that prints the username and follower count.



```
#include <iostream>
#include <string>
using namespace std;

class SocialMediaUser {
public:
    string username;
    int followers;

    // Constructor
    SocialMediaUser(string username, int followers) {
        this->username = username;
        this->followers = followers;
    }

    // Method to display profile
    void displayProfile() {
        cout << "Username: " << username << endl;
        cout << "Followers: " << followers << endl;
    }
};

int main() {
    SocialMediaUser user("Haaris", 10000);

    user.displayProfile();

    return 0;
}
```

OUTPUT:

```
Username: Haaris  
Followers: 10000
```

2. Build a YouTuber class that inherits from SocialMediaUser and adds a property channelName and a method uploadVideo(title) that prints 'Video [title] uploaded to [channelName]'.



```
class YouTuber : public SocialMediaUser {  
public:  
    string channelName;  
  
    // Constructor  
    YouTuber(string username, int followers, string channelName)  
        : SocialMediaUser(username, followers) {  
        this->channelName = channelName;  
    }  
  
    // Upload video method  
    void uploadVideo(string title) {  
        cout << "Video " << title  
            << " uploaded to " << channelName << endl;  
    }  
};  
  
int main() {  
    YouTuber yt("Haaris", 10000, "Haaris Tech");  
  
    yt.displayProfile();  
    yt.uploadVideo("C++ Tutorial");  
  
    return 0;  
}
```

OUTPUT:

```
Username: Haaris  
Followers: 10000  
Video C++ Tutorial uploaded to Haaris Tech
```

3. Create a Podcaster class that also inherits from SocialMediaUser and adds a property podcastName and a method publishEpisode(episodeTitle) that prints 'Episode [episodeTitle] published on [podcastName]'.



```
class Podcaster : public SocialMediaUser {
public:
    string podcastName;

    // Constructor
    Podcaster(string username, int followers, string podcastName)
        : SocialMediaUser(username, followers) {
        this->podcastName = podcastName;
    }

    // Publish episode method
    void publishEpisode(string episodeTitle) {
        cout << "Episode " << episodeTitle
            << " published on " << podcastName << endl;
    }
};

int main() {
    Podcaster pod("Haaris", 5000, "Tech Talks");

    pod.displayProfile();
    pod.publishEpisode("Episode 1");

    return 0;
}
```

OUTPUT:

```
Username: Haaris
Followers: 5000
Episode Episode 1 published on Tech Talks
```

4. Demonstrate multilevel inheritance by creating a class GamingYouTuber that inherits from YouTuber and adds a method streamGame(gameName) which prints '[username] is now streaming [gameName] on [channelName]'.
→

```
class GamingYouTuber : public YouTuber {
public:

    GamingYouTuber(string username, int followers, string channelName)
        : YouTuber(username, followers, channelName) {

    }

    void streamGame(string gameName) {
        cout << username
            << " is now streaming "
            << gameName
            << " on "
            << channelName << endl;
    }
};

int main() {
    GamingYouTuber gamer("HaarisGaming", 20000, "Haaris Gaming");

    gamer.displayProfile();
    gamer.uploadVideo("Gaming Highlights");
    gamer.streamGame("GTA V");

    return 0;
}
```

OUTPUT:

```
Username: HaarisGaming
Followers: 20000
Video Gaming Highlights uploaded to Haaris Gaming
HaarisGaming is now streaming GTA V on Haaris Gaming
```

5. Implement hierarchical inheritance by creating an InstagramInfluencer class that inherits from SocialMediaUser and adds a method postStory(storyTitle) which prints '[username] posted a new story: [storyTitle]'.
Hint: Think about how SocialMediaUser is the parent for YouTuber, Podcaster, and InstagramInfluencer.



```
#include <iostream>
#include <string>
using namespace std;

class SocialMediaUser {
public:
    string username;
    int followers;

    SocialMediaUser(string username, int followers) {
        this->username = username;
        this->followers = followers;
    }

    void displayProfile() {
        cout << "Username: " << username << endl;
        cout << "Followers: " << followers << endl;
    }
};

// InstagramInfluencer inherits from SocialMediaUser
class InstagramInfluencer : public SocialMediaUser {
public:

    InstagramInfluencer(string username, int followers)
        : SocialMediaUser(username, followers) {
    }

    void postStory(string storyTitle) {
        cout << username
            << " posted a new story: "
            << storyTitle << endl;
    }
};

int main() {
    InstagramInfluencer influencer("HaarisOfficial", 15000);

    influencer.displayProfile();
    influencer.postStory("My New Setup");
}
```

```
    return 0;  
}
```

OUTPUT:

```
Username: HaarisOfficial  
Followers: 15000  
HaarisOfficial posted a new story: My New Setup
```